

WEEKLY PROGRESS REPORT – WEEK 2

Project Title

Healthcare Analytics Pipeline for MediCare Hospital

Group Number: 3

Team Members

- Arpit Sharma (G25AI1012)
 - Ankita Dwivedi (G25AI1011)
 - Astosh Ranjan (G25AI1013)
 - Bharat Sharma (G25AI1014)
 - Damini Khatri (G25AI1015)
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Week 2 Objectives

The objective of Week 2 was to clean, preprocess, validate, and standardize the healthcare dataset to ensure data quality and consistency for further processing and analysis.

Activities Completed

1. Data Cleaning and Preprocessing

The healthcare dataset was reviewed to identify missing values, inconsistent records, duplicate entries, and formatting issues. Data cleaning techniques were applied to improve overall dataset quality.

2. Missing Value Handling

Missing values were identified and handled using appropriate preprocessing techniques. Null values and incomplete records were analyzed to ensure data consistency.

3. Data Validation

Validation checks were performed to verify data accuracy, completeness, and consistency across different dataset attributes.

4. Data Standardization

Dataset fields were standardized to maintain uniform formatting and improve compatibility with downstream processing systems.

5. Data Privacy and Compliance Review

The dataset was reviewed to ensure that no personally identifiable or sensitive healthcare information was exposed. Data handling practices were aligned with ethical and privacy requirements.

6. Data Dictionary Preparation

A data dictionary was created to document all dataset attributes, their descriptions, data types, and intended usage within the project.

7. Data Lineage Documentation

Data lineage documentation was prepared to track the flow of data from source acquisition through cleaning and preprocessing stages.

8. Team Coordination and Review

The team reviewed the cleaned dataset, validated preprocessing results, and discussed the next phase of ETL pipeline implementation.

Deliverables Completed

- Cleaned healthcare dataset
- Missing value handling documentation
- Data validation report
- Standardized dataset
- Data dictionary
- Data lineage documentation
- Data quality review

Challenges Faced

- Handling incomplete records
- Ensuring data consistency across attributes
- Standardizing healthcare data formats
- Maintaining privacy and data quality requirements

Plan for Week 3

- Develop ETL pipeline
 - Integrate AWS S3 storage
 - Configure PostgreSQL database
 - Automate workflow using Apache Airflow
 - Validate data loading process
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Conclusion

Week 2 focused on data cleaning, preprocessing, validation, and documentation. The healthcare dataset was successfully prepared for ETL processing and subsequent integration into the data engineering pipeline.